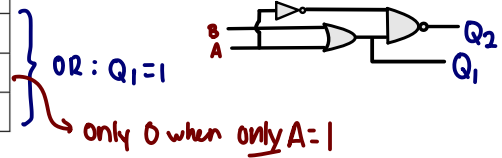


Assignment 10

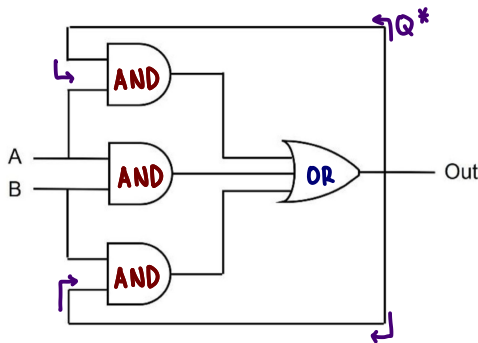
Due Friday, April 24, 2026

1) Design a circuit using basic logic gates that will implement the following truth table:

Input A	Input B	Output 1	Output 2
0	0	0	1
0	1	1	1
1	0	1	0
1	1	1	1



2) Describe the behavior of the circuit below. (Hint: consider what output states are self-consistent with each combination of 0 or 1 applied to the inputs A and B). Note that the circuit element on the right is a 3-input OR gate. Its output is high if at least one of the inputs is high.



$$Q^* = AQ + BQ + AB$$

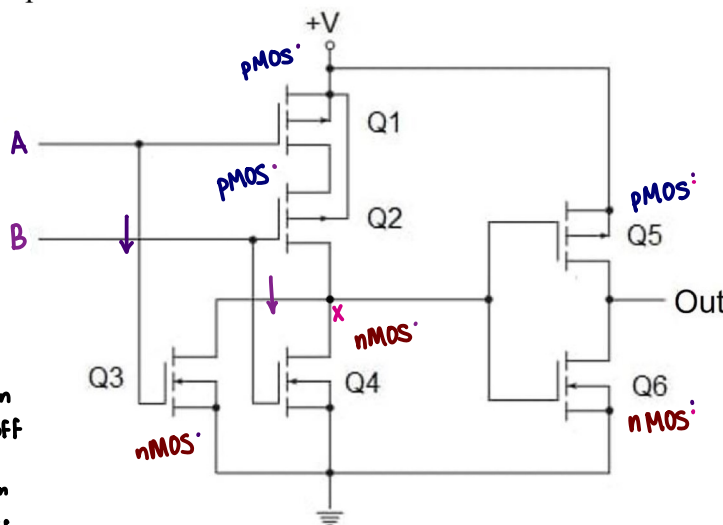
$$Q = Q(A+B) + A+B$$

Input		Output
A	B	
0	0	0 → sets Q* to 0
1	0	1 or 0: depends on value of Q*
0	1	1 or 0: depends on value of Q*
1	1	1 → sets Q* to 1

depends on 'stored value of Q*' from immediately prior calculation such that $Q = Q^*$

This circuit exhibits the behavior of self-referential memory and utilizes said memory in it's logic.

3) Write out a truth table for the CMOS logic gate that is shown below. What logic operation does this correspond to?



A	B	output
0	0	0
1	0	1
0	1	1
1	1	1

A controls Q₁, Q₃
B controls Q₂, Q₄
Q₂, Q₃, Q₄ control X { where Q₁, Q₂ → pullup and Q₃, Q₄ → nMOS pulldown
X = A + B → inverted by Q₅, Q₆
L = A + B
X controls Q₅, Q₆

⇒ Behaves like OR Gate

nMOS $V_{gs} = V_g - V_s$
pMOS $V_{gs} = V_s - V_g$
g = 1: on
g = 0: off

for $V_{in} = +5V$
pMOS $V_{gs} = 0V$ off
nMOS $V_{gs} = 5V$ on

for $V_{in} = 0V$
pMOS $V_{gs} = -5V$ on
nMOS $V_{gs} = 0V$ off

OR $A \text{ OR } B \rightarrow A+B$

Input		Output
A	B	
0	0	0
1	0	1
0	1	1
1	1	1